OAKG AAAI 2027

Experimental Guidelines for UD

Observability-Aware Knowledge Graph Reasoning for Heterogeneous Medical Imaging Data

Objective

Complete the new experiments needed to establish that OAKG provides benefits beyond its internal coverage-blind ablation and beyond simple missing-data handling. All final results must be reproducible, leakage-free, and directly usable in the AAAI 2027 paper.

Companion notebook: OAKG_AAAI2027_Experiment_Notebook.ipynb

Prepared July 18, 2026

1. Immediate Priorities

Highest-priority acceptance test

OAKG must be compared directly with masked cosine under the independent full-annotation partial-observation benchmark. The key question is whether OAKG improves retrieval beyond simply restricting pairwise comparison to jointly observed features.

Priority	Task	Required result	Status
P0	Strong coverage-aware baselines	Masked cosine, missingness indicators, Gower, zero and mean imputation	Required
P0	Independent partial-observation benchmark	Reference-derived and predicted-segmentation tracks; four masking regimes	Required
P0	Ranking-policy selection	S, gamma*S, threshold+S, and lexicographic policies	Required
P0	Paired statistics	95% paired bootstrap CIs; Holm correction	Required
P1	Selective retrieval	Risk-coverage curve, served rate, AURC	Strongly recommended
P1	Graph and CT baselines	WL, CompGCN, observed-region CT, hybrid methods	Strongly recommended
P1	Cross-backbone observability	Base vs. +Obs under identical input evidence	Strongly recommended
P2	Structured semantics and diagnostics	Closed/open/OAKG reasoning; query and pool statistics	Required for completeness

2. Non-Negotiable Experimental Controls

- Use patient-level train, validation, and test splits. No masked realization of a patient may appear in a different split.
- Fix all phenotype thresholds, model hyperparameters, ranking policies, alpha values, and evidence thresholds using only training and validation data.
- Use identical queries, candidate pools, relevance labels, masking realizations, and seeds across directly compared methods.
- Do not use reference masks to correct predicted segmentations or predicted phenotypes in the end-to-end track.
- Reference annotations may define method-independent relevance labels and the controlled image-access boundary, but may not supply reference-derived semantic features to end-to-end graph or phenotype models.
- Report the full-volume CT encoder separately as a full-information upper bound. Do not include it in OAKG-vs-baseline deltas.
- Retain all query-level predictions and scores so every aggregate metric can be reproduced.

3. Two Required Input Tracks

Track	Retrieval inputs	Fixed evaluation target	Purpose
Controlled mechanism	Masked reference-derived phenotypes, graphs, and observed-region CT input	Binary and graded relevance from complete reference annotations	Isolate the missing-evidence mechanism without segmentation error
End-to-end	Masked predicted-segmentation phenotypes and graphs; same observed-region CT boundary	The same complete-reference relevance labels	Measure observability together with segmentation and phenotype-extraction error

$$\text{Delta\_upstream}^{(b)} = \text{Metric\_ref}^{(b)} - \text{Metric\_pred}^{(b)}$$

Report this degradation separately for each method, metric, and masking condition. Do not pool the two tracks.

4. Masking Regimes

Regime	Implementation	Main interpretation
Uniform	Query and all candidates retain the same organ subset	Matched-evidence control
Random	Approximately 20%, 40%, 60%, and 80% missing anatomical coverage; multiple seeds	Performance as evidence decreases
Dataset-style	Pancreas-only, liver-only, kidney-only, and multi-organ patterns	Realistic source-style coverage
Asymmetric	Broad-to-narrow and narrow-to-broad masking	Direction-specific retrieval difficulty

Important: Each query and candidate must use its own observation mask. Record the exact organ set retained for every case and every seed.

5. Strong Missing-Data Baselines

- Zero-imputed phenotype vector: replace unobserved values with zero; rank by cosine similarity.
- Mean/mode-imputed phenotype vector: use training-set means for numerical features and modes for categorical features.
- Phenotype vector with missingness indicators: concatenate phenotype values with a binary observation mask.
- Masked cosine: compare only features observed in both cases. This is the most important simple baseline.
- Gower similarity: compute mixed numerical/categorical similarity while excluding dimensions missing for either case.

Required reporting: Precision@10, Recall@10, mAP, nDCG@10, and served-query rate in aggregate and by matched, cross-dataset, decomposition, masking level, direction, and input track.

Critical paper comparison

Report OAKG minus masked cosine with a paired 95% confidence interval. This comparison determines whether the graph-based observability mechanism adds value beyond pairwise feature masking.

6. Ranking-Policy Ablation and Final OAKG Definition

Policy	Definition	Interpretation
Similarity only	R = S	Coverage is reported but does not alter ranking
Product	R = gamma * S	Continuously weights similarity by shared evidence
Thresholded	Eligible when gamma >= gamma_min; rank by S	Uses coverage as an eligibility condition
Lexicographic	Evidence category first, then S	Separates evidence strength from phenotype similarity

- Select the primary policy using validation nDCG@10 together with served-query rate.
- Fix the selected policy before evaluating the test set.
- Confirm which policy generated the existing OAKG results. The earlier results appear to use gamma*S.
- If validation selects another policy, rerun the principal OAKG tables using the selected policy. Retain gamma*S as an ablation.

7. Selective Retrieval and Abstention

Risk(gamma_min) = 1 - nDCG@10(gamma_min), computed over served queries

- Vary gamma_min over a validation-defined grid.
- For every threshold report served-query rate, nDCG@10 over served queries, selective risk, and the area under the risk-coverage curve.
- Plot risk versus served-query coverage for OAKG and the strongest coverage-aware baselines.
- Never report improved nDCG without the corresponding served-query rate; high performance obtained by excessive abstention must be visible.

8. Graph, CT, and Hybrid Baselines

8.1 Weisfeiler-Lehman graph kernel

- Construct the case-centered graph using exactly the nodes, relation labels, and available evidence permitted by the masking condition.
- Report kernel depth, node-label construction, normalization, and any ontology labels included.

8.2 CompGCN relational graph encoder

- Use one fixed implementation, not an optional CompGCN/R-GCN description.
- Record the training objective, positive and negative pair construction, dimensions, layers, composition operator, pooling, optimizer, learning rate, epochs, weight decay, and random seeds.
- Select model configuration using validation queries only and export case embeddings for reproducible retrieval evaluation.

8.3 Observed-region CT embedding

- Replace voxels outside the permitted observation region with one fixed background value.
- Pool encoder features only over the observed anatomical region.
- Use the same observed-region image input for the CT baseline and every hybrid method.
- Record the exact encoder architecture, checkpoint, pretraining dataset, image spacing, crop size, intensity normalization, frozen/fine-tuned status, and pooling method.
- Report a full-volume CT encoder separately as a full-information upper bound.

8.4 Hybrid image-graph retrieval

$$S_{hybrid} = \alpha \cdot S_{graph} + (1 - \alpha) \cdot S_{CT\_obs}$$

- Normalize graph and CT similarities using validation-set statistics before combination.
- Select alpha on validation data only.
- Coverage-blind hybrid: observed-region CT score plus coverage-blind graph score.
- Observability-aware hybrid: the same CT score plus support-restricted graph similarity and the selected eligibility policy.

9. Cross-Backbone Observability Test

$$\Delta_{obs}^{(b,t)} = \mathrm{Metric}_{(b+\mathrm{Obs},t)} - \mathrm{Metric}_{(b,t)}, \quad t \in \{\mathrm{ref}, \mathrm{pred}\}$$

Evaluate this for phenotype vectors, WL, CompGCN, observed-region CT embeddings, and hybrid image-graph retrieval.

- The base and +Obs variants must use the same architecture, parameters, input source, and permitted evidence.
- For phenotype and graph backbones, +Obs removes unsupported elements from pairwise comparison and applies OAKG comparability/eligibility.
- For CT, both variants use observed-region embeddings; +Obs adds pair-level comparability and the selected evidence policy.
- Exclude the full-volume CT upper bound from Delta_obs.

10. Ranking Consistency and Structured Reasoning

10.1 Full-to-partial ranking consistency

- Compare each method only with its own complete-input ranking from the same input track.
- Report Kendall tau, Spearman correlation, and top-10 overlap.
- Do not compare a partial predicted-input ranking with a complete reference-derived ranking.

10.2 Structured-query semantics

- Compare closed-world reasoning, generic open-world reasoning, and OAKG support-aware three-valued reasoning.

- Report the three-class confusion matrix for T/F/U, per-class precision, recall, F1, macro-F1, indeterminate rate, and unsupported-negative rate.
- Verify that a failed prediction inside retained anatomy remains a prediction error and is not relabeled as unobserved.

11. Query Diagnostics and Statistical Analysis

11.1 Query and candidate diagnostics

- Number of queries by phenotype type and coverage/masking stratum.
- Candidate-pool size for each query.
- Mean, median, minimum, and maximum number of relevant candidates.
- Relevance prevalence and number of zero-relevant queries.
- Counts of broad-to-narrow and narrow-to-broad queries.
- Number and percentage of unserved queries by method and condition.

11.2 Statistical protocol

- Use paired bootstrap resampling over queries with 10,000 repetitions for the final study.
- For random masking, incorporate both query resampling and masking-seed variation.
- Report 95% confidence intervals for absolute metrics and paired method differences.
- Apply Holm correction within each primary comparison family.
- Emphasize effect sizes and confidence intervals, not p-values alone.

12. Required Paper-Ready Outputs

Output	Purpose	Required content
Table A	Strong baselines	P@10, R@10, mAP, nDCG@10, served rate; aggregate and three original strata
Table B	Full-annotation masking	Reference and predicted tracks across masking levels
Table C	Ranking policies	S, gamma*S, threshold, lexicographic; selected policy identified
Table D	Cross-backbone observability	Base vs. +Obs deltas for each backbone and track
Table E	Ranking consistency	Kendall, Spearman, top-10 overlap
Table F	Structured semantics	T/F/U metrics and unsupported-negative rate
Figure 1	Risk-coverage	Selective risk versus served-query coverage
Figure 2	Masking stress	nDCG@10 versus missing-coverage level for key methods

13. Files and Deliverables to Return

- Executed notebook using real data, with no hidden manual steps.
- query_level_retrieval_results.csv
- retrieval_summary.csv
- upstream_degradation.csv
- risk_coverage_curve.csv and the corresponding figure
- ranking_consistency.csv
- structured_query_predictions.csv and structured_query_summary.csv
- query_diagnostics.csv
- publication_ready_retrieval_table.csv
- CompGCN and CT embedding files with case IDs and embeddings
- Configuration files containing all thresholds, seeds, data splits, checkpoints, and preprocessing settings
- A short README describing how to reproduce every table and figure

Recommended folder structure

```
oakg_aaai2027/
  data_schema/
  configs/
  notebooks/
  src/
  embeddings/
  results/query_level/
  results/tables/
  results/figures/
  checkpoints/
  README.md
```

14. Final Submission Checklist

☐ All new experiments run on the fixed test split after validation choices are frozen.

☐ OAKG versus masked cosine includes paired confidence intervals.

☐ Reference-derived and predicted-segmentation results are reported separately.

☐ The final OAKG ranking policy is explicitly identified.

☐ Full-volume CT is labeled as an upper bound and excluded from direct deltas.

☐ All cross-backbone comparisons hold permitted evidence constant.

☐ Every table includes the number of evaluated and served queries.

☐ All TBD values in the manuscript are replaced by measured results or removed.

☐ All scripts, seeds, configurations, and output files are archived.

☐ The revised Abstract, Results, Discussion, and Conclusion use only verified final values.

Definition of completion

The work is complete only when the final tables can be regenerated from the saved query-level outputs and the selected validation configuration without manual editing.